

Mathematical Logic - unit 1

The Rules of Logic are used in the design of computer circuits and the construction of computer program, the verification of programs and many other ways.

Proposition:

A proposition is a declarative sentence that is either true (or) false but not both.

Every proposition associated to two values i.e, true / 1 (or) false / 0.

Ex: 1) Mango is a fruit - True / 1.

2) $\sqrt{5}$ is a Prime number - False / 0.

Logical Operations:

1) Negation: ($\neg$ (or) $\sim$ (or) $-$)

A proposition obtained by inserting the word "not" at an appropriate place in a given proposition is called Negation.

Suppose 'P' be a given proposition then negation of 'P' is denoted by

$\sim P$ / $\neg P$ / $\bar{P}$.

Ex: p : Today is Sunday.

$\sim p$: Today is not Sunday.

Here ' $\sim$ ' is called negation operator.

2) Conjunction ($\wedge$): A Proposition obtained by two given propositions by inserting the word 'and' in between them is called Conjunction, which is true only when both P and Q are true. Otherwise it is false. Conjunction of P and Q is denoted by $P \wedge Q$.

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

3) Disjunction ($\vee$): A Proposition obtained by two given propositions by inserting the word 'or' in between them is called Disjunction, which is false only when both P and Q are false, otherwise it is true. Disjunction of P and Q is $P \vee Q$.

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

- 4) Conditional Statement ($\rightarrow$): A Compound proposition obtained by combining two given propositions by using the words "if" and "then" at appropriate place is called Conditional Statement.

Let P and Q be two propositions, the conditional Statement ($P \rightarrow Q$) is a proposition, which is false only when if P is true and Q is false, otherwise it is true.

P	Q	$P \rightarrow Q$
T	T	T
T	F	F
F	T	T
F	F	T

Here P is called Hypothesis and Q is called Conclusion.

Note:

- 1) The converse of $P \rightarrow Q$ is $Q \rightarrow P$.
- 2) The Inverse of $P \rightarrow Q$ is $\sim P \rightarrow \sim Q$.
- 3) Contra positive of $P \rightarrow Q$ is $\sim Q \rightarrow \sim P$.

5) Biconditional Statements ($\leftrightarrow$): A compound proposition obtained by combining by given two propositions by using the words "if and only if" at appropriate place is called Biconditional Statements.

The Biconditional of P and Q is denoted by $P \leftrightarrow Q$, which is true only when both truth values are same. Otherwise it is false.

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

6) Exclusive OR ($\oplus$):

Suppose P and Q be two propositions, the Exclusive OR of P and Q is denoted by $P \oplus Q$, which is true when the exactly one of P and Q is true, otherwise it is false.

P	Q	$P \oplus Q$
T	T	F
T	F	T
F	T	T
F	F	F

Problems:

Q) Construct the truth table for following propositions

i) $P \wedge (\sim Q)$

Sol:

P	Q	$\sim Q$	$P \wedge (\sim Q)$
T	T	F	F
T	F	T	T
F	T	F	F
F	F	T	F

ii) $\neg(\sim P) \vee Q$

P	Q	$\sim P$	$(\sim P) \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

iii) $Q \rightarrow \sim P$

P	Q	$\sim P$	$Q \rightarrow \sim P$
T	T	F	F
T	F	F	T
F	T	T	T
F	F	T	T

iv) $P \oplus \sim Q$

P	Q	$\sim Q$	$P \oplus \sim Q$
T	T	F	T
T	F	T	F
F	T	F	F
F	F	T	T

$$v) (P \wedge Q) \rightarrow (P \vee Q)$$

P	Q	$P \wedge Q$	$P \vee Q$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	F

$P \wedge Q$	$P \vee Q$	$(P \wedge Q) \rightarrow (P \vee Q)$
T	T	T
F	T	T
F	T	T
F	F	T

$$vi) (Q \rightarrow \neg P) \leftrightarrow (P \rightarrow Q)$$

$$vii) P \oplus P \vee Q$$

$$viii) (P \vee \neg Q) \rightarrow (P \wedge Q)$$

$$9) (P \leftrightarrow Q) \oplus (\neg P \leftrightarrow \neg Q)$$

$$10) (P \leftrightarrow Q) \oplus (\neg P \leftrightarrow Q)$$

vi)

P	Q	$\neg P$	$Q \rightarrow \neg P$	$P \rightarrow Q$	$(Q \rightarrow \neg P) \leftrightarrow (P \rightarrow Q)$
T	T	F	F	T	F
T	F	F	T	F	F
F	T	T	T	T	T
F	F	T	T	T	T

vii)

P	Q	$P \vee Q$	$P \oplus P \vee Q$
T	T	T	F
T	F	T	F
F	T	T	T
F	F	F	F

10) $(P \leftrightarrow Q) \oplus (\sim P \leftrightarrow Q)$

$(Q \vee \sim Q)$

P	Q	$\sim P$	$P \leftrightarrow Q$	$\sim P \leftrightarrow Q$	$(P \leftrightarrow Q) \oplus (\sim P \leftrightarrow Q)$
T	T	F	T	F	T
T	F	F	F	T	T
F	T	T	F	T	T
F	F	T	T	F	T

Write Converse, inverse and Contrapositive for the following stmt's:

- 1) Converse $P \rightarrow Q = Q \rightarrow P$
- 2) inverse $P \rightarrow Q = \sim P \rightarrow \sim Q$
- 3) Contra +ve $P \rightarrow Q = \sim Q \rightarrow \sim P$

- 1) P : 2 is a real number
 Q : 9 is multiple of 3

a) Converse of $P \rightarrow Q$ is $Q \rightarrow P$

i.e. if 2 is a real number then 9 is a multiple of 3.

If 9 is a multiple of 3 then 2 is a real number.

b) Inverse of $P \rightarrow Q$ is $\sim P \rightarrow \sim Q$

if 2 is not a real number then 9 is not a multiple of 3.

c) Contradictive of $P \rightarrow Q$ is $\sim Q \rightarrow \sim P$

If 9 is not a multiple of 3 then 2 is not a real number.

2) P : Today is a holiday

Q : I will go for movie.

a) Converse of $P \rightarrow Q$ is $Q \rightarrow P$

If today is a holiday then I will go for movie.

If I will go for movie then Today is a holiday.

b) Inverse of $P \rightarrow Q$ is $\sim P \rightarrow \sim Q$

If Today is not a holiday then I will not go for movie.

c) Contradictive of $P \rightarrow Q$ is $\sim Q \rightarrow \sim P$

I will not go for movie then today is not a holiday.

3) If $\triangle ABC$ is a right angle then $AC^2 = AB^2 + BC^2$

sol: Let P : $\triangle ABC$ is a right angle Δ .

Q : $AC^2 = AB^2 + BC^2$

Converse of $P \rightarrow Q$ is $Q \rightarrow P$

If $AC^2 = AB^2 + BC^2$ then $\triangle ABC$ is a right angle Δ .

Contradictive of $p \rightarrow q$ is $\neg q \rightarrow \neg p$
If $AC^2 \neq AB^2 + BC^2$ then $\triangle ABC$ is not right angled

Inverse of $p \rightarrow q$ is $\neg p \rightarrow \neg q$
If $\triangle ABC$ is not a right angled then $AC^2 \neq AB^2 + BC^2$

4) Write the following stmts in symbolic form.

p : Mark is happy.

q : Mark is rich.

a) Mark is happy and rich $\Rightarrow p \wedge q$

b) Mark is poor (or) he is both rich and happy.

c) Mark is neither rich nor happy. $\neg q \vee (\neg p)$

d) Mark is poor but happy $\Rightarrow \neg q \wedge p$

5) If John takes computer Science (or) Mercy takes Mech then Nishant takes civil.

Write the above stmt in symbolic form.

Sol: Let p : John takes CSE

q : Mercy takes Mech

r : Nishant takes civil

$$(p \vee q) \rightarrow r$$

6) Write the Symbolic form for following Stmts.
 p : It is below freezing
 q : It is Snowing

a) It is below freezing & Snowing $\Rightarrow p \wedge q$

b) It is below freezing & not Snowing $\Rightarrow p \wedge \sim q$

c) It is not below freezing but not Snowing.
 $\sim p \wedge \sim q$

d) It is either Snowing (or) below freezing.
 $q \vee p$

e) If it is below freezing then it is Snowing.
 $p \rightarrow q$

f) It is either below freezing (or) it is Snowing but if it is not Snowing then it is below freezing. $\Rightarrow (p \vee q) \wedge (\sim q \rightarrow p)$

g) It is below freezing if necessary & Sufficient condition for it to be Snowing. $p \leftrightarrow q$.

7) Suppose P & q be 2 propositions given by

P : Swimming at new jersey Shore is allowed

q : Sharks have been spotted near the shore.

Express the following compound propositions as and English Sentences.

- a) $\sim q, p \wedge q$ c) $\sim q \rightarrow p \wedge \sim p \wedge (p \vee \sim q)$
 b) $p \rightarrow \sim q$ d) $\sim p \rightarrow \sim q$

a) $\sim q \Rightarrow$ Sharks have not been spotted near the shore
 $p \wedge q \Rightarrow$ Swimming at New J is allowed and sharks have been spotted near the shore.

b) $p \rightarrow \sim q \Rightarrow$ If swimming at N.J is allowed then sharks have not been spotted.

c) $\sim q \rightarrow p \Rightarrow$ If sharks have not been spotted near the shore then swimming at N.J is allowed.

d) $\sim p \rightarrow \sim q \Rightarrow$ If swimming at N.J is not allowed then sharks have not been spotted.

e) $\sim p \wedge (p \vee \sim q) \Rightarrow$ Swimming at N.J is not allowed but swimming at N.J is allowed (or) sharks have not been spotted near the shore.

Statement formula :

A Statement formula is an Expression denoted by a string consisting of variable connecting symbols and parenthesis.

Well formed Formula (WFF):

A Grammatically correct Expression is called a WFF. A WFF can be generated by the following rules:

- 1) A Statement variable's & constants is a WFF.
- 2) If P is a WFF then $\sim P$ is a WFF.
- 3) If P & Q are variable's then $P \wedge Q$, $P \vee Q$, $P \rightarrow Q$ --- is a WFF.
- 4) A String of Symbols containing the Statement variables, connectives and parenthesis is a WFF if & only if it can be obtained by finite applications of rules of above 1, 2, 3.

Ex: 1). $P \rightarrow (P \wedge Q)$ is a WFF.

2) $(P \vee Q) \rightarrow (P \wedge (P \wedge \sim Q))$ is a WFF.

3) $P \rightarrow P \wedge (P \vee \sim Q)$ not a WFF because there is no parenthesis.

4) $(P \wedge Q) \rightarrow P \vee Q$ is (not) a WFF.

Logically Equivalent:

A compound proposition that have same truth values in all possible cases are called Logically Equivalent and it is denoted by ' $\equiv$ ' (or) ' $\cong$ '.

Q) $A = (p \rightarrow q) \vee (p \rightarrow r)$ & $B = p \rightarrow (q \vee r)$

P	q	r	$p \rightarrow q$	$p \rightarrow r$	A	$q \vee r$	B
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

Q) Show $\sim(p \vee q)$ & $\sim p \wedge \sim q$

P	q	$p \vee q$	$\sim(p \vee q)$	$\sim p$	$\sim q$	$\sim p \wedge \sim q$
T	T	T	F	F	F	F
T	F	T	F	F	T	F
F	T	T	F	T	F	F
F	F	F	T	T	T	T

Q) $A = (p \rightarrow (q \rightarrow r))$; $B = (p \rightarrow (\sim q \wedge r))$

check $C = (p \wedge q) \rightarrow r$ are Logically Equivalent or not.

Sol:

P	q	r	$q \rightarrow r$	A	$\sim q$	$\sim q \wedge r$	B	$P \wedge q$	C
T	T	T	T	T	F	F	F	T	T
T	T	F	F	F	F	F	F	T	F
T	F	T	T	T	T	T	T	F	T
T	F	F	F	F	T	F	F	F	T
F	T	T	T	T	F	F	T	F	T
F	T	F	F	T	F	F	T	F	T
F	F	T	T	T	T	T	T	F	T
F	F	F	T	T	T	F	T	F	T

$$\therefore A \neq B \text{ \& } A \equiv C$$

Q) S.T the following Stmt's are logically Equivalent.

Sol: Let P : Good food is not cheap

q : cheap food is not good

The given Statements in Symbolic form

$$P \rightarrow \sim q \text{ and } q \rightarrow \sim P$$

P	q	$\sim P$	$\sim q$	$P \rightarrow \sim q$	$q \rightarrow \sim P$
T	T	F	F	F	F
T	F	F	T	T	T
F	T	T	F	T	T
F	F	T	T	T	T

$$\therefore P \rightarrow \sim q \equiv q \rightarrow \sim P$$

Q) Determine $\neg[(p \wedge q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)] \equiv A$
is a tautology or not.

P	q	r	$p \wedge q$	$p \wedge q \rightarrow r$	$q \vee r$	$p \rightarrow (q \vee r)$	A
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	F	F
F	T	T	F	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	F	T	T

Q) $\neg(p \vee q) \vee (\neg p \wedge q) \vee p \equiv A$

P	q	$p \vee q$	$\neg(p \vee q)$	$\neg p$	$\neg p \wedge q$	B	A
T	T	T	F	F	F	F	T
T	F	T	F	F	F	F	T
F	T	T	F	T	T	T	T
F	F	F	T	T	F	T	T

$\therefore A$ is a Tautology.

Propositional Equivalents:

1) Tautology: (All True's in the final column)

A compound Proposition is always true is called a tautology.

2) Contradiction: A compound proposition is always false.

3) Contingency: A compound proposition is neither tautology nor contradiction is called contingency.

Q) 8.T $[P \wedge (P \rightarrow Q)] \rightarrow P \equiv A$

P	Q	$P \rightarrow Q$	$P \wedge (P \rightarrow Q)$	$[P \wedge (P \rightarrow Q)] \rightarrow P$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

$\therefore A$ is a tautology.

Q) $A \equiv (P \wedge \sim Q) \wedge Q$ is a contradiction. Prove

P	Q	$\sim Q$	$P \wedge \sim Q$	A
T	T	F	F	F
T	F	T	T	F
F	T	F	F	F
F	F	T	F	F

$\therefore A$ is a contradiction.

Q) $[p \rightarrow (q \vee r)] \leftrightarrow [(p \rightarrow q) \vee (p \rightarrow r)] = A \leftrightarrow B$

P	q	r	$q \vee r$	A	$p \rightarrow q$	$p \rightarrow r$	B	$A \leftrightarrow B$
T	T	T	T	T	T	T	T	T
T	T	F	T	T	T	F	T	T
T	F	T	T	T	F	T	T	T
T	F	F	F	F	F	F	F	T
F	T	T	T	T	T	T	T	T
F	T	F	T	T	T	T	T	T
F	F	T	T	T	T	T	T	T
F	F	F	F	T	T	T	T	T

$\therefore A \leftrightarrow B$ is a Tautology.

Q) Show that $(p \wedge q) \rightarrow (p \vee q)$ is a tautology.

Sol:

Let $A \equiv (p \wedge q) \rightarrow (p \vee q)$

Q) Show that $\sim(p \rightarrow q) \rightarrow \sim q$ is a tautology.

$$\begin{aligned}\text{Let } A &\equiv \sim(p \rightarrow q) \rightarrow \sim q \\ &\equiv \sim(\sim(p \rightarrow q)) \vee \sim q \\ &\equiv (p \rightarrow q) \vee \sim q \\ &\equiv (\sim p \vee q) \vee \sim q \\ &\equiv \sim p \vee (q \vee \sim q) \\ &\equiv \sim p \vee T \equiv T\end{aligned}$$

$\therefore A$ is tautology.

a) Show that $(p \wedge q) \rightarrow (p \rightarrow q)$ is a tautology.

$$\begin{aligned}\text{Let } A &\equiv (p \wedge q) \rightarrow (p \rightarrow q) \\ &\equiv \sim(p \wedge q) \vee (p \rightarrow q) \\ &\equiv \sim(p \wedge q) \vee (\sim p \vee q) \\ &\equiv (\sim p \vee \sim q) \vee (\sim p \vee q) \\ &\equiv (\sim p \vee \sim p) \vee (\sim q \vee q)\end{aligned}$$

$$\equiv F \vee T \equiv T$$

$\therefore A$ is a tautology.

Duality Law: Two formulas A & B are said to be duals of each other if one can be obtained by other by replacing ' $\wedge$ ' by ' $\vee$ ' & ' $\vee$ ' by ' $\wedge$ '.

Normal forms: The construction of truth values may not be practicable even with the aid of computer. Therefore, we consider other procedure reduction to normal forms in this method we use the word product in place of ' $\wedge$ ' and sum in place of ' $\vee$ '.

Elementary Sum: The sum of the variables and their negation in a formula is called Elementary sum. Ex: Let P, Q be two variables then their elementary sum is Ex: $P \vee Q, P \vee \neg P, P \vee \neg Q, \neg P \vee P, \dots$

Elementary Product: The product of the variables and their negation in a formula is called Elementary product. Ex: Let P, Q, R be variables then their elementary product is $P \wedge Q \wedge R, P \wedge \neg Q \wedge R, \neg P \wedge \neg Q \wedge R, \dots$

Minterm: It consists of conjunction in which each variable or its negation but not both and appears only once is called

Ex: Let p, q, r be three variables then its min term is $p \wedge q \wedge r, \neg p \wedge q \wedge r, \neg p \wedge \neg q \wedge r, \dots$

Max term: It consists of disjunction in which each variable or its negation but not both and appears only once is called max term.

Ex: Let p, q, r be three var then, $p \vee q \vee r, \neg p \vee \neg q \vee \neg r, \neg p \vee q \vee \neg r, \dots$

Conjunctive normal forms: A formula which is Equivalent to given formula and consists of product of Elementary Sum is called Conjunctive normal form. Ex: Let p, q be 2 vars then $(p \vee q) \wedge (\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (q \vee p).$

Disjunctive normal form (DNF): A formula in which is Equivalent to given formula is and consists of Sum to Elementary product is called disjunctive normal form. Let p, q be 2 variables then its DNF is $(p \wedge q) \vee (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (q \wedge p).$

Principle conjunctive normal forms: A formula which is Equivalent to given formula and consists max terms is called PCNF.
(rules) $() \wedge () \wedge ().$

Principle disjunctive normal form: A formula which is equivalent to given formula and consists of min terms is called PDNF (read) $() \vee () \vee ()$.

Working Procedure to write PDNF:

$$A \equiv (P \wedge q) \rightarrow [P \vee (\sim q \vee P)]$$

Step-1: Construct the truth table for the given Stmt formula.

Step-2: Select the truth value T corresponding to the Stmt formula.

Step-3: Write the sum of min terms to the corresponding truth values as follows

- i) If truth value of P is T then take it as P.
- ii) If truth value of P is F then it as negation ($\sim P$).

Q) Obtain PDNF of $(P \leftrightarrow q) \oplus (P \rightarrow \sim q) \equiv A$

P	q	$P \leftrightarrow q$	$\sim q$	$P \rightarrow \sim q$	A
T	T	T	F	F	T
T	F	F	T	T	T
F	T	F	F	F T	T
F	F	T	T	T	F

$$\text{PDNF} = (P \wedge q) \vee (P \wedge \sim q) \vee (\sim P \wedge q)$$

False x, for True only.

Working Procedure to write PCNF:

(rules)
Step-1: Construct the TT for given stmt formula.

Step-2: Select the truth value F corresponding to the stmt formula.

Step-3: Write the product of max terms to the corresponding truth values as follows

i) If truth value of P is T then it is $\sim p$.

ii) If truth value of P is F then it is p .

a) Obtain PCNF and PDNF of $p \wedge (p \leftrightarrow q)$

p	q	$p \leftrightarrow q$	$p \wedge (p \leftrightarrow q)$
T	T	T	T
T	F	F	F
F	T	F	F
F	F	T	F

PCNF

$$(\sim p \vee q) \wedge (p \vee \sim q) \wedge (p \vee q)$$

PDNF

$$(p \wedge q)$$

a) Obtain PCNF & PDNF of

$$p \vee (\sim p \wedge \sim q) \vee \neg \equiv A.$$

$$5. \left. \begin{aligned} (p \vee q) \vee r &= p \vee (q \vee r) \\ (p \wedge q) \wedge r &= p \wedge (q \wedge r) \end{aligned} \right\} \text{Associative law}$$

$$6. \left. \begin{aligned} (p \vee q) \wedge r &= (p \wedge r) \vee (q \wedge r) \\ p \vee (q \wedge r) &= (p \vee q) \wedge (p \vee r) \end{aligned} \right\} \text{Distributive law}$$

$$7. \left. \begin{aligned} \neg(p \vee q) &= \neg p \vee \neg q \\ \neg(p \wedge q) &= \neg p \vee \neg q \end{aligned} \right\} \text{De-Morgan's law}$$

$$8. \left. \begin{aligned} p \vee (p \wedge q) &= p \\ p \wedge (p \vee q) &= p \end{aligned} \right\} \text{Absorption law} \quad \left[\begin{aligned} (p \wedge q) \vee q &= q \\ (p \vee q) \wedge q &= q \end{aligned} \right]$$

$$9. \left. \begin{aligned} p \vee \neg p &= T \\ p \wedge \neg p &= F \end{aligned} \right\} \text{Negation law}$$

Note :

$$1. p \rightarrow q \equiv \sim p \vee q$$

$$2. p \rightarrow q \equiv \sim q \rightarrow \sim p$$

$$3. p \vee q \equiv \neg(\neg p \wedge \neg q) \quad \sim p \rightarrow q$$

$$4. p \wedge q \equiv \neg(p \rightarrow \sim q)$$

$$5. \neg(p \rightarrow q) \equiv p \wedge \sim q$$

$$6. (p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$$

$$7. (p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$$

$$8. (p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$$

$$9. (p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$$

p	q	r	$\sim p$	$\sim q$	$\sim p \wedge \sim q \wedge r$	A
T	T	T	F	F	F	T
T	T	F	F	F	F	T
T	F	T	F	T	F	T
T	F	F	F	T	F	T
F	T	T	T	F	F	F
F	T	F	T	F	F	F
F	F	T	T	T	T	T
F	F	F	T	T	F	F

$$\text{PDNF} = ((p \wedge q) \wedge r) \vee (p \wedge q \wedge \sim r) \vee (p \wedge \sim q \wedge r) \vee (p \wedge \sim q \wedge \sim r) \vee (\sim p \wedge \sim q \wedge r)$$

$$\text{PCNF} = (p \vee \sim q \vee \sim r) \wedge (p \vee \sim q \vee r) \wedge (p \vee q \vee r)$$

Logical Equivalence formula:

- | <u>Formula</u> | <u>Name</u> |
|---|-----------------|
| 1. $p \wedge p = p$
$p \vee p = p$ | Identity law |
| 2. $p \vee T = T$
$p \wedge F = F$ | |
| 3. $\neg(\neg p) = p$ | Double negation |
| 4. $p \vee q = q \vee p$
$p \wedge q = q \wedge p$ | commutative law |

5. $(p \vee q) \wedge (p \wedge q)$

6. $(p \vee q) \wedge p$

7. $\neg(\neg p)$

8. $p \vee (p \wedge q)$

9. $p \wedge (p \vee q)$

Note:

1. $p \vee (p \wedge q) = p$

2. $p \wedge (p \vee q) = p$

3. $p \vee (p \wedge q) = p$

4. $p \wedge (p \vee q) = p$

5. $p \vee (p \wedge q) = p$

6. $p \wedge (p \vee q) = p$

7. $p \vee (p \wedge q) = p$

8. $p \wedge (p \vee q) = p$

9. $p \vee (p \wedge q) = p$

$$10. (p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$11. (p \leftrightarrow q) \equiv (\sim p \leftrightarrow \sim q)$$

$$12. (p \leftrightarrow q) \equiv (p \wedge q) \vee (\sim p \wedge \sim q)$$

$$13. \sim(p \leftrightarrow q) \equiv p \leftrightarrow \sim q$$

Obtain PCNF and PDNF of $A \equiv \neg p \vee q$

Sol:

$$\text{Let } A \equiv \neg p \vee q$$

$$\equiv (\neg p \wedge \top) \vee (q \wedge \top)$$

$$\equiv (\neg p \wedge (q \wedge \neg q)) \vee (q \wedge (p \vee \neg p))$$

$$\text{PDNF} \equiv (\neg p \wedge q) \vee (\neg p \wedge \neg q) \vee (q \wedge p) \vee (q \wedge \neg p)$$

$$\equiv (\neg p \vee q) \text{ PCNF}$$

Q) Obtain PCNF and PDNF of $p \wedge q$

Sol:

$$\text{Let } A \equiv p \wedge q$$

$$\text{PDNF} \equiv (p \wedge q)$$

$$\text{PCNF} = A \equiv p \wedge q$$

$$\equiv (p \vee \neg p) \wedge (q \vee \neg q)$$

$$\equiv (p \vee (q \wedge \neg q)) \wedge (q \vee (p \wedge \neg p)) \quad [\text{Negation law}]$$

$$\equiv (p \vee q) \wedge (p \vee \neg q) \wedge (q \vee p) \wedge (q \vee \neg p)$$

$$\equiv (p \vee q) \wedge (p \vee \neg q) \wedge (q \vee \neg p)$$

Q) Obtain PCNF and PDNF of $A \equiv \neg(p \rightarrow q)$

Sol:

$$A \equiv \neg(p \rightarrow q)$$

$$\equiv \neg(\neg p \vee q)$$

$$\equiv \neg(\neg p) \wedge \neg q$$

$$\begin{array}{l} 8) \quad P \vee q \\ \quad \quad \neg P \vee r \\ \hline \therefore q \vee r \end{array}$$

$$(P \vee q) \wedge (\neg P \vee r) \rightarrow q \vee r \quad \text{resolution}$$

Check the validity of the following arguments

- 1) "If there were an Examination then catching by way difficulties."
- 2) "If they arrived on Time then catching by way difficulties."
- 3) "They arrived on Time ; There was no Explanation."

Let P = There was an Examination

$\sim P$ = There was no Examination

q = catching blw was difficult

r = Arrived on Time

$\sim q$ = catching the bus was not difficult

Q) "If it rains today then we will not have holiday today." ~~today~~

"If we don't have holiday today then we have holiday tomorrow."

"Therefore, if it rains today then we will have a holiday tomorrow."

Now $PCNF = A \equiv p \wedge \neg q$
 $\equiv (p \vee F) \wedge (\neg q \vee F)$

Q) Obtain PCNF and PDNF of $\neg(p \vee q)$

Sol: Let $A \equiv (\neg p \wedge \neg q)$

$PDNF \equiv (\neg p \wedge \neg q)$

$PCNF \equiv (\neg p \vee F) \wedge (\neg q \vee F)$

$= (\neg p \vee (q \wedge \neg q)) \wedge (\neg q \vee (p \wedge \neg p))$

$= (\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg q \vee p) \wedge (\neg q \vee \neg p)$

$= (\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (\neg q \vee \neg p) \wedge (\neg q \vee p)$

Theory of Inference:

The main function of the logic is to provide rules of inference, This theory is associated with such rules is known as Theory of Inference.

Argument: An argument is a group of statements including one (or) more premises and one conclusion.

Premises: Premise is an axiom (or) believed to be through Either from Experience (or) faith.

Conclusion: A conclusion is a Statement in a argument, that indicates what the arguments is trying to convince the reader.

Rules of Inference

Tautology

Name

- | | | |
|---|--|------------------------|
| 1) $\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$ | $p \wedge (p \rightarrow q) \rightarrow p$ | Modus ponens |
| 2) $\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$ | $(p \rightarrow q) \wedge (q \rightarrow r) \rightarrow (p \rightarrow r)$ | Hypothetical Syllogism |
| 3) $\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \neg p \end{array}$ | $\neg q \wedge (p \rightarrow q) \rightarrow \neg p$ | Modus Tollens |
| 4) $\begin{array}{l} p \vee q \\ \neg p \\ \hline \therefore q \end{array}$ | $(p \vee q) \wedge \neg p \rightarrow q$ | Disjunctive Syllogism |
| 5) $\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$ | $p \rightarrow p \vee q$ | Addition |
| 6) $\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$ | $(p \wedge q) \rightarrow p$ | Simplification |
| 7) $\begin{array}{l} q \\ \hline \therefore p \wedge q \end{array}$ | $p \wedge q \rightarrow (p \wedge q)$ | Simplification |

Sol: p : It rains today.

q : We will have holiday today.

r : We will have holiday tomorrow.

In Symbolic form, $p \rightarrow \sim q$
 $\sim q \rightarrow r$

 $\therefore p \rightarrow r$

Verification:

1) $p \rightarrow \sim q$ Premise

2) $\sim q \rightarrow r$ Premise

3) $p \rightarrow r$

Q) "If you send me an Email message then I will finish writing the program".

"If you do not send an e-mail msg then I will go to sleep early".

"If I go to sleep early then I will wake up feeling refresh".

"Therefore, If I don't finish writing the program then I will wake up feeling refresh".

Sol: Let P : You send me an e-mail msg.
 q : I will finish writing the program.
 r : I will go to sleep early.
 s : I will wake up feeling refresh.

In Symbolic form, $P \rightarrow q$
 $\sim P \rightarrow r$
 $r \rightarrow s$

 $\therefore \sim q \rightarrow s$

Verification:

1) $P \rightarrow q$ Premise

2) $\sim P \rightarrow r$ Premise

3) $r \rightarrow s$ Premise

4) $\sim q \rightarrow \sim P$ by contrative of $(P \rightarrow q)$ ①

5) $\sim q \rightarrow r$ by hypothetic syl of ④ & ③

6) $\sim q \rightarrow s$ by hypo syl of ⑤ & ③

 $\therefore \sim q \rightarrow s$

a) "If clifton does not live in france then he does not speak french".

"clifton does not drive a datson".

"If clifton lives in france then he ~~writes~~ rides a bicycle."

"Either clifton speak french or he drives a datson".

The validity of the argument by using Truth Tables:

"If in an argument, Let $H_1, H_2, \dots, H_n$ be the premises & 'c' be the conclusion.

If $(H_1 \wedge H_2 \wedge \dots \wedge H_n) \rightarrow c$ is a Tautology then the argument is valid otherwise it is invalid".

Check the validity of argument by using T.T:

$$P \rightarrow Q$$

$$\frac{\sim(P \wedge Q)}{\therefore \sim P}$$

$$\text{Let } [(P \rightarrow Q) \wedge (\sim(P \wedge Q))] \equiv A$$

P	Q	$P \rightarrow Q$	$\sim P \wedge Q$	$\sim(P \wedge Q)$	A	$A \rightarrow \sim P$
T	T	T	F	F	F	T
T	F	F	T	T	F	T
F	T	T	F	T	T	F
F	F	T	F	T	T	T

"Hence clifton drives a bicycle".

Sol: p : clifton lives in France.

q : clifton speak French.

r : clifton drives a datson.

s : He rides a bicycle.

In Symbolic form, $\sim p \rightarrow \sim q$
 $\sim r$

$p \rightarrow s$

$q \vee r$

$\therefore s$

Verification:
= = = = =

1) $\sim p \rightarrow \sim q$

Premise

2) $\sim r$

Premise

3) $p \rightarrow s$

"

4) $q \vee r$

"

5) $q \rightarrow p$

by Contradictive of (1)

6) $q \rightarrow s$

by Hyp. syl of (3) & (5)

7) q

by hyp syl of (2) & (4)

8) s

by modus ponens of (6) & (7)

$\therefore s$

In Symbolic form, $\forall x (A(x) \rightarrow B(x))$
 $\forall x (D(x) \rightarrow \sim C(x))$
 $\forall x (B(x) \rightarrow C(x))$

 $\therefore \forall x (A(x) \rightarrow \sim D(x))$

Verification:

1) $\forall x (A(x) \rightarrow B(x))$

2) $\forall x (D(x) \rightarrow \sim C(x))$

3) $\forall x (B(x) \rightarrow C(x))$

4) $A(s) \rightarrow B(s)$

5) $D(s) \rightarrow \sim C(s)$

6) $B(s) \rightarrow C(s)$

7) $C(s) \rightarrow \sim D(s)$

8) $A(s) \rightarrow C(s)$

$A(s) \rightarrow \sim D(s)$

by contrative of (5)

by hypsil of (1) & (6)

" " " of (7) & (8)

Indirect method of Proof:

The notation of inconsistent is used in a procedure is called indirect method of proof.

In order to show that a conclusion C logically follows from the premises $H_1, H_2, \dots, H_n$. We assume that conclusion C is false and consider $\sim C$ as an additional premise.

If the new set of premises is inconsistent then the assumed premise $\sim C$ is false

Q) check the validity of the arguments for the following by using quantifiers.

1) "All integers are rational numbers"

"Some integers are powers of 2"

"Therefore Some rational numbers are powers of 2".

Sol:

Let $A(x)$: x is an integer

$B(x)$: x is a rational

$C(x)$: x is a power of 2

In Symbolic form,

$$\frac{\forall x (A(x) \rightarrow B(x)) \quad \exists x (A(x) \wedge C(x))}{\therefore \exists x (B(x) \wedge C(x))}$$

Verification:

1) $\forall x (A(x) \rightarrow B(x))$

Premise

2) $\exists x (A(x) \wedge C(x))$

"

3) $A(s) \rightarrow B(s)$

by rule of US of ①

4) $A(s) \wedge C(s)$

by rule of ES of ②

5) $A(s)$

Simplification of ④

6) $C(s)$

Simplification of ④

7) $B(s)$

by mod. ponens of ③ & ⑤

8) $B(s) \wedge C(s)$

by conjunction of ⑥ & ⑦

9) $\exists x (B(x) \wedge C(x))$ by rule of EG of ⑧

$\therefore \exists x (B(x) \wedge C(x))$

2) "Babies are illogical".

"Nobody is despised who can manage a crocodile".

"Illogical people are despised"

Therefore babies cannot manage a crocodile.

Sol:

Let

$A(x)$: x is a baby

$B(x)$: x is illogical

$C(x)$: x can be despised

$D(x)$: x can manage a crocodile

Consistency: A set of formulas $H_1, H_2, \dots, H_n$ is said to be consistent if their conjunction has the truth value T for some assignment of truth values to the atomic variable appearing in the premises.

Inconsistent: A set of formulas $H_1, H_2, \dots, H_n$ is said to be inconsistent if their conjunction has the truth value F.

Q) Show that following premises are inconsistent.

Sol:

1) $P \rightarrow q$

Premise

2) $P \rightarrow r$

"

3) $q \rightarrow \sim r$

"

4) P

"

5) $\sim q \rightarrow \sim p$

by contrative of (1)

6) q

by m.p of (1) & (4)

7) $\sim r$

by m.p of (3) & (6)

8) $\sim p$

by m.p of (2) & (4)

9) $P \wedge \sim p$

by conjunction of (4) & (8)

$$\frac{P \wedge \sim p}{F}$$

$\therefore$ The given arguments are inconsistent